Lesson plan · for the teacher

How a chatbot writes

AI · Level 2 · Lesson 1

Time
40 min

Materials
Board and chalk; A starter for each round, such as “In the rainy season, farmers…”, and one nobody can know, such as “Our school was opened in the year…”; The textbook page on the topic (to check facts)

Learning goals

- Build a sentence one likely word at a time, like a chatbot.
- Explain why a sentence can sound right and still be untrue.
- Check a fact against the textbook.

Activity: Human chatbot

1. Warm-up (5 min): Write “I will see you…” on the board and ask pupils what word comes next. Collect three ideas and ask which is most likely. Explain that a phone keyboard does the same: it suggests a likely next word.
2. Teach (8 min): Explain that a chatbot writes the same way, one small piece at a time, choosing a likely next one from patterns in huge amounts of text. Stress that it does not check whether its sentence is true.
3. Human chatbot (15 min): Write a starter on the board, such as “In the rainy season, farmers…”. Six pupils stand in a line; each adds one likely word until the sentence ends. Play three rounds with different starters. Make one a starter nobody can know, such as “Our school was opened in the year…”: pupils still give a likely-sounding year, just as a chatbot invents a detail. In another round, let a pupil pick an unlikely word, and see how the sentence changes.
4. Check it (7 min): Read each finished sentence aloud. The class votes: true, false or not sure. For each one, ask how we could check: the textbook, the teacher, an elder. Check one together in the textbook.
5. Wrap-up (5 min): Ask: how was our game like a chatbot? Agree that sounding right is not the same as being true. Then give the exit ticket.

Prepare three starters: two from this week’s topics, for example “In the rainy season, farmers…” and “The heart pumps…”, and one nobody can know, such as “Our school was opened in the year…”. Have the textbook page ready, so the class can check at least one sentence.

Discussion

- Why did some of our sentences sound right even when they weren’t true?
- When is it fun that a chatbot picks words with a little chance, and when is it a problem?
- How could we check a fact that a chatbot gives us?

Exit ticket

- Write one sentence that sounds right but is not true, and how you would check it.

Try it at school tomorrow
Play “Human chatbot”: Pupils build a sentence together, one word each, each choosing a likely next word. Read the finished sentence aloud and ask: does it sound right? Is it true? How would we check?

Practice questions · for you

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1 **Amina asks a chatbot for a rain story, then asks again in a new chat with exactly the same words. She gets a different story. Why?**

- ◯ Its training changed between her questions
- ◯ It picks likely tokens, with a little chance each time
- ◯ It copied another story from the web

2 **Which of these can make a chatbot's answer wrong, even when it sounds sure? Choose all that apply.**

- ☐ Its training text ends before recent events
- ☐ You asked in a language it saw little text in
- ☐ It predicts likely words instead of looking facts up
- ☐ It invents a detail when it doesn't know
- ☐ It is lying to you on purpose

3 **The same message usually takes the same number of tokens in every language.**

- ◯ True
- ◯ False

4 **Put in order: how a chatbot writes its reply.**

- ☐ Score every possible next token
- ☐ Cut your message into tokens
- ☐ Repeat until the reply is finished
- ☐ Pick a likely token, with a little chance
- ☐ Add it to the reply

5 **Match each surprise to its cause inside the chatbot.**

It doesn't know last month's exam results	• • • •	More tokens for the same message
A reply in Hausa is slower and uses up a free limit sooner	• • • •	Its knowledge cut-off
The same request gives a new poem each time	• • • •	Invented details that fit the pattern
It names a textbook that doesn't exist	• • • •	A little chance in each pick

✎ **Think of something your pupils might ask a chatbot. Where could a fluent answer mislead them, and how would you teach them to check?**

Check your answers

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1 **It picks likely tokens, with a little chance each time**

At each step the model picks among likely next tokens, with a little chance built in, so the same request can produce a new story.

2 **It predicts likely words instead of looking facts up; Its training text ends before recent events; It invents a detail when it doesn't know; You asked in a language it saw little text in**

Prediction without checking, a knowledge cut-off, invented details and a language with little text online can all make a fluent answer wrong.

3 **False**

Tokens were learned mostly from English text, so the same message in Arabic, Hausa or Yoruba is often cut into more tokens, sometimes several times as many: slower, costlier, and limits come sooner.

4 **Top to bottom: 2, 1, 5, 3, 4**

Your message becomes tokens; then, for each new token, the model scores the options, picks a likely one, adds it and goes round again.

5 **It doesn't know last month's exam results — Its knowledge cut-off**
A reply in Hausa is slower and uses up a free limit sooner — More tokens for the same message
The same request gives a new poem each time — A little chance in each pick
It names a textbook that doesn't exist — Invented details that fit the pattern

Each everyday surprise has a cause inside the chatbot: a cut-off date, token counts, a little chance, and prediction without checking.